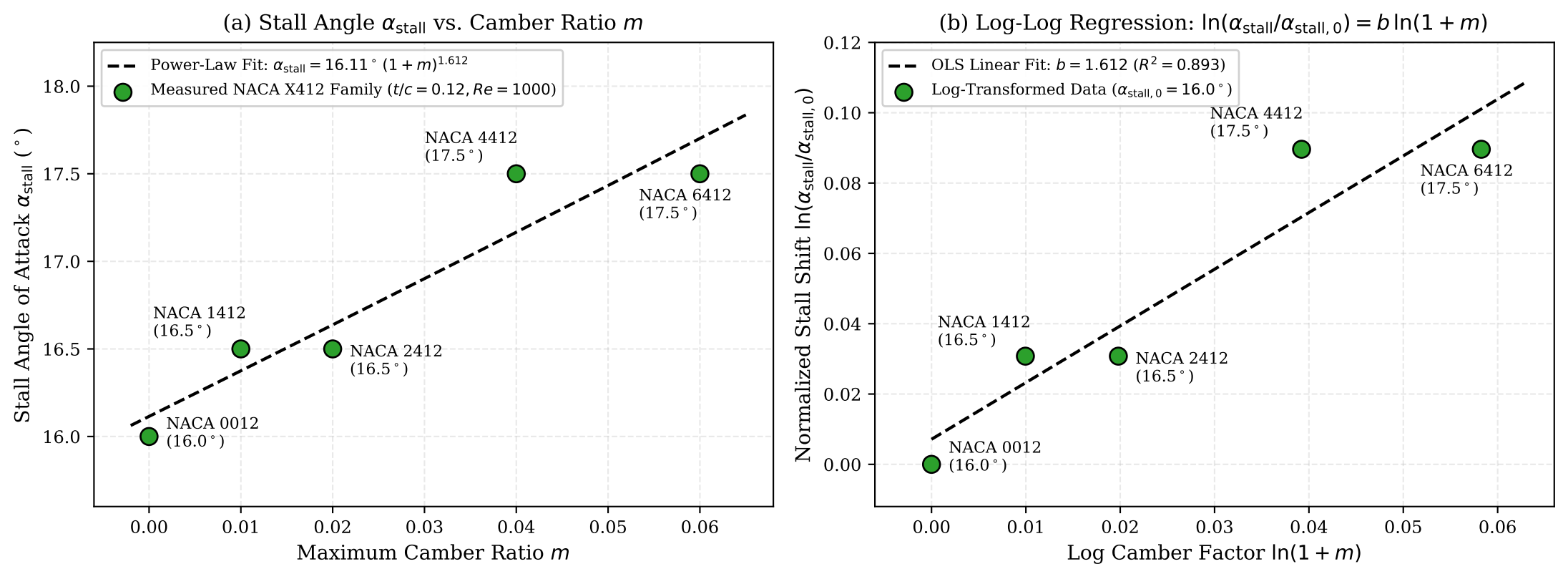
(a) Stall Angle α_{stall} vs. Camber Ratio m
Power-Law Fit: $\alpha_{stall} = 16.11^\circ\,(1+m)^{1.612}$
Measured NACA X412 Family ($t/c = 0.12, Re = 1000$)
NACA 4412 (17.5°)
NACA 6412 (17.5°)
NACA 1412 (16.5°)
NACA 2412 (16.5°)
NACA 0012 (16.0°)
Stall Angle of Attack α_{stall} (°)
Maximum Camber Ratio m
18.0
17.5
17.0
16.5
16.0
0.00
0.01
0.02
0.03
0.04
0.05
0.06
(b) Log-Log Regression: $\ln(\alpha_{stall}/\alpha_{stall,0}) = b\ln(1+m)$
OLS Linear Fit: $b = 1.612$ ($R^2 = 0.893$)
Log-Transformed Data ($\alpha_{stall,0} = 16.0^\circ$)
NACA 4412 (17.5°)
NACA 6412 (17.5°)
NACA 1412 (16.5°)
NACA 2412 (16.5°)
NACA 0012 (16.0°)
Normalized Stall Shift $\ln(\alpha_{stall}/\alpha_{stall,0})$
Log Camber Factor $\ln(1+m)$
0.12
0.10
0.08
0.06
0.04
0.02
0.00
0.00
0.01
0.02
0.03
0.04
0.05
0.06